

Recursive Program for Fibonacci numbers in C:

The fibonacci sequence(series) is the sequence of integers 0, 1, 1, 2, 3, 5, 8, 13, 21, 34,

Each item in this series is the sum of the two preceding elements (for example, $0+1 = 1$, $1+1 = 2$, $1+2 = 3$, $2+3 = 5$,)

If we let $\text{fib}(0) = 0$, $\text{fib}(1) = 1$, and so on then we may define the fibonacci series by the following recursive definition:

$$\text{Fib}(n) = n \text{ if } n=0 \text{ or } n=1$$

$$\text{Fib}(n) = \text{fib}(n-2) + \text{fib}(n-1) \text{ if } n \geq 2$$

Example,

```
Int fib( int n)
{
int x, y;
```

```
if(n==0 || n==1) return (n);
```

```
else
```

```
{
```

```
  X = fib(n-1);
```

```
  Y=fib(n-2);
```

```
  return(x+y);
```

```
}
```